# Energy-Based Models

Learn an energy landscape that assigns favorable scores to real observations, then use an iterative sampler to explore low-energy regions without calculating the normalizing constant.

Original study notes based on David Foster, Second edition, 2023. Book pp. 189–204. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

Real and sampled images → Energy network → Contrastive objective → Input-gradient sampling → Replay and refine

## Detailed Notes

### Energy and the Boltzmann distribution

Source: book pp. 189–191 / PDF p. 217

An energy-based model assigns a scalar E(x) to each observation. Low energy corresponds to higher probability after normalization. The partition function sums/integrates over all possible observations and is generally intractable for image spaces.

> $p_\theta(x) = \exp(-E_\theta(x)) / Z_\theta$
> $Z_\theta = \int \exp(-E_\theta(x))\, dx$

Some implementations output a score equal to negative energy. That reverses gradient directions and score ordering, so establish the convention before interpreting losses. An unrestricted energy function is flexible, but sampling and normalization remain hard.

### MNIST and the energy function

Source: book pp. 192–193 / PDF p. 220

The MNIST project uses a convolutional network mapping an image to a single scalar, rather than a probability for each digit. Real digits define favorable regions and generated negative examples teach what to reject.

Normalize the input domain and do not add a sigmoid merely because another chapter used binary classification. Suitable activations and stable score scales matter. The gradient with respect to image pixels is needed during sampling, while the gradient with respect to network weights is needed during learning; these are different derivatives.

## Langevin dynamics sampling

Source: book pp. 194–196 / PDF p. 222

Start from noise or previous samples, repeatedly add a stochastic perturbation and follow the input gradient toward lower energy. Under a common convention:

> $x_{k+1} = x_k - (\eta/2)\nabla_x E(x_k) + \sqrt{\eta}\,\varepsilon_k$

Practical implementations may use different step/noise scaling and clipping. The noise encourages exploration; gradient descent without it is not the same sampling procedure. Finite steps and poor mixing bias the distribution. Check intermediate images, gradient magnitudes, and the behavior of pixel-range constraints.

## Contrastive divergence and replay

Source: book pp. 197–200 / PDF p. 225

Likelihood gradients compare real-data energies with model-sampled energies. Contrastive divergence approximates the otherwise expensive negative-sampling term using a limited sampling run. Replay buffers reuse previously generated examples, reducing repeated initialization and encouraging more mature negative samples.

> Approximate contrastive objective = E[E(real)] - E[E(negative)]

Minimize this expression under the low-energy-real convention, with appropriate score regularization. Samples are ordinarily treated as fixed negatives during the parameter update rather than differentiating through the whole sampler. Approximate negative samples make the procedure different from exact maximum-likelihood training.

## Analysis, limitations, and other energy models

Source: book pp. 201–204 / PDF p. 229

Visualize the sampling trajectory, compare real/generated energy distributions, and test whether the network has learned useful distinctions or merely increased score magnitude. Realistic finite-step samples do not demonstrate that the sampler has mixed across all modes.

Classical Boltzmann machines and restricted Boltzmann machines use related energy formulations, often with hidden variables and Gibbs sampling. Deep image EBMs use neural energies and gradient-based sampling. Key challenges include sampler cost, replay quality, instability, and interpreting unnormalized scores as probabilities.

---

# Revision Check

1. Why is the partition function difficult?
2. How does score=-energy change the sampler sign?
3. What does Langevin noise add beyond gradient descent?
4. Why is contrastive divergence approximate?

5. How can a replay buffer help and also hide poor exploration?

> For EBMs, the sampler is part of the model's practical behavior. Inspect its mixing and finite-step bias as carefully as the energy network.

---

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.